RAG-Powered Knowledge Agents 1: Embeddings, Chunking & Indexing

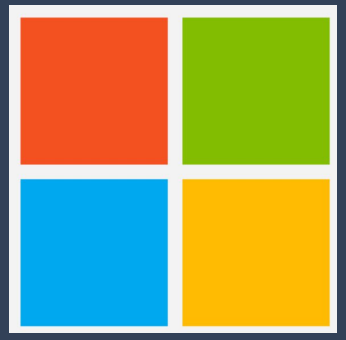
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Introduction - Jeevendra Singh



- I have been working as a Software Engineer since 2013
- I've worked for companies like Ericsson, SAP, Q2, and now Microsoft
- I have been working at Microsoft for 4+ years
- In my career, I've worked as a full-stack developer, integration architect, and now focusing on Agentic AI development
- Built intelligent API platforms and copilots using Generative AI, RAG, and Agent Orchestration

Currently: **Developing AI copilots with Microsoft Copilot Studio, RAG, and Agent Orchestration for Business and Industry Copilot (BIC) team**







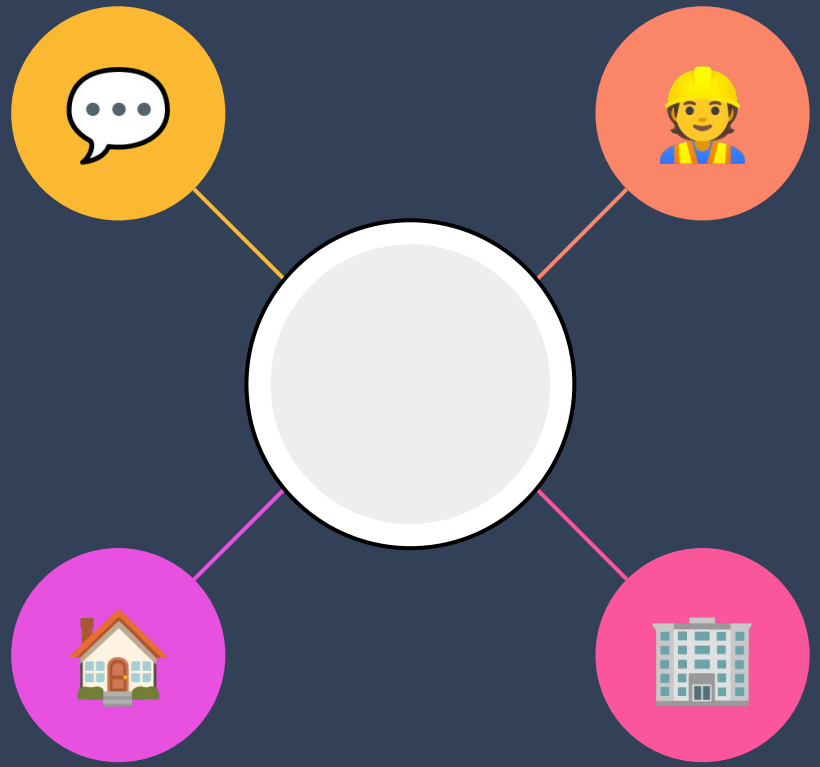


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Welcome to today's class, before we begin, pop into the chat:

Hannah Chen

Data Scientist



Bay Area

Grammarly, ex-Uber

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Structure of Class

1. **Sunday Class - 4 hr (Sunday)**
2. **Assignments -** MCQ & Coding Assignment





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Optimize Your Experience



- ✅ Interact with your instructors via Sunday live class.
- ✅ Don’t be shy to speak up and get clarifications !



- ✅ Solve your assignments, MCQs and other assessments and get feedback (Thursday review session)
- ✅ Use your resources! It’s your experience – what you put in is what you’ll get out.

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Don’t worry!

We are here to help at your every step !

But, how???



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Are you using all these IK support features?

- **Uplevel** - contains all the learning resources - videos, MCQs, assignments
- Optional **Post class videos** for additional reference
- **Technical coaching sessions** on Wednesday to clarify all your queries
- **Discord** community for peer learning
- **TAs** to help answer the queries in the live class
- **Support tickets** - Raise them any time you want any clarifications/ tech concerns , etc on UpLevel

Success Hacks

- Be prepared for the class by watching the pre class videos
- Attend every class
- Participate in class
- Attempt all the assignments,
- Actively advocate for yourself
- Keep your goal in mind
- Consistency is the key

Be patient with yourself !

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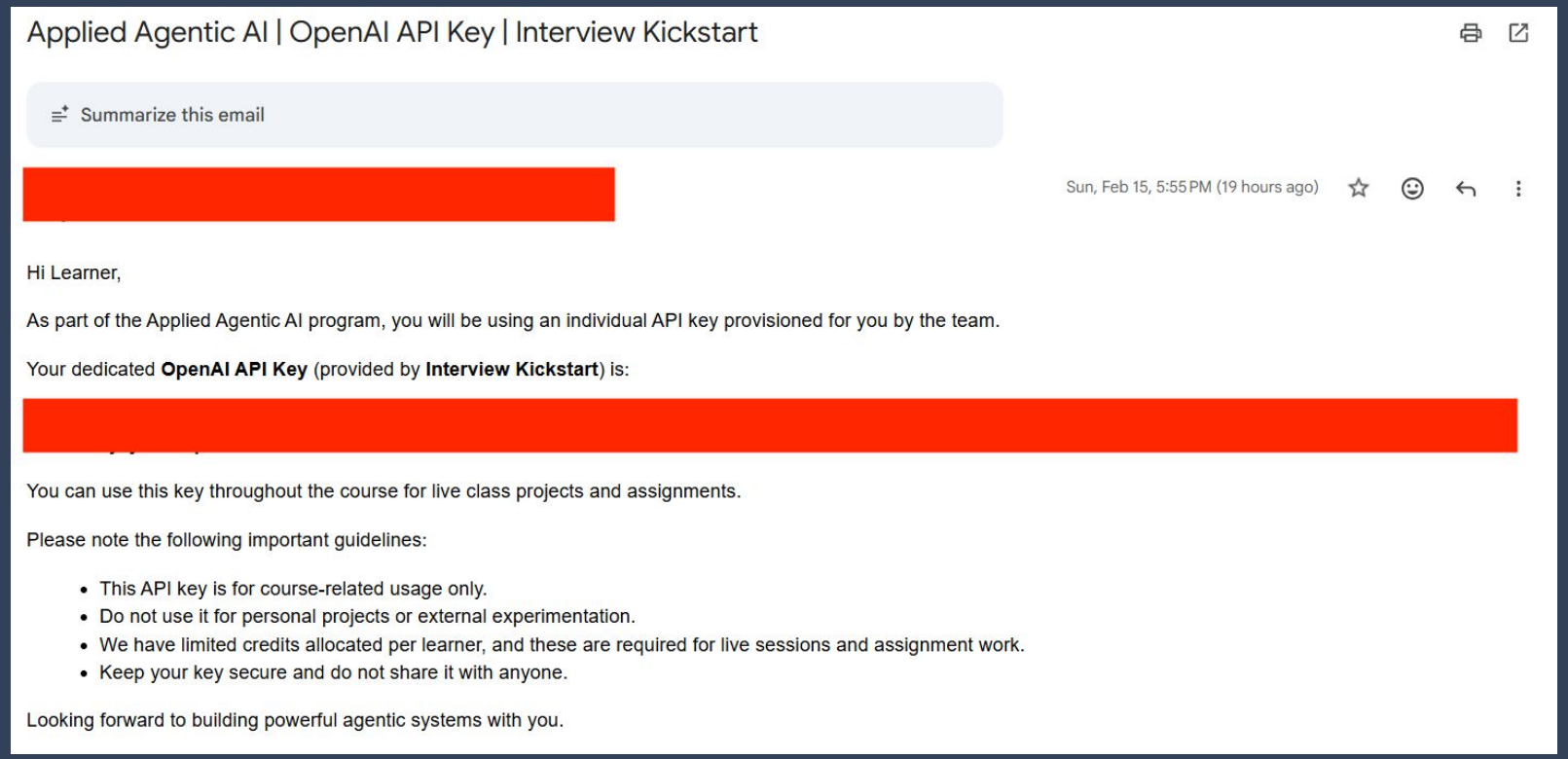
Check Your Email for API Key

You should have received an email like the one shown below containing your OpenAI API key.

If you do not see this email:

- Please check your inbox
- Also check your spam or junk folder

If you still cannot find it, please reach out to the Operations Team operations@interviewkickstart.com



Applied Agentic AI | OpenAI API Key | Interview Kickstart

Summarize this email

[redacted]

Sun, Feb 15, 5:55 PM (19 hours ago)

Hi Learner,

As part of the Applied Agentic AI program, you will be using an individual API key provisioned for you by the team.

Your dedicated **OpenAI API Key** (provided by **Interview Kickstart**) is:

[redacted]

You can use this key throughout the course for live class projects and assignments.

Please note the following important guidelines:

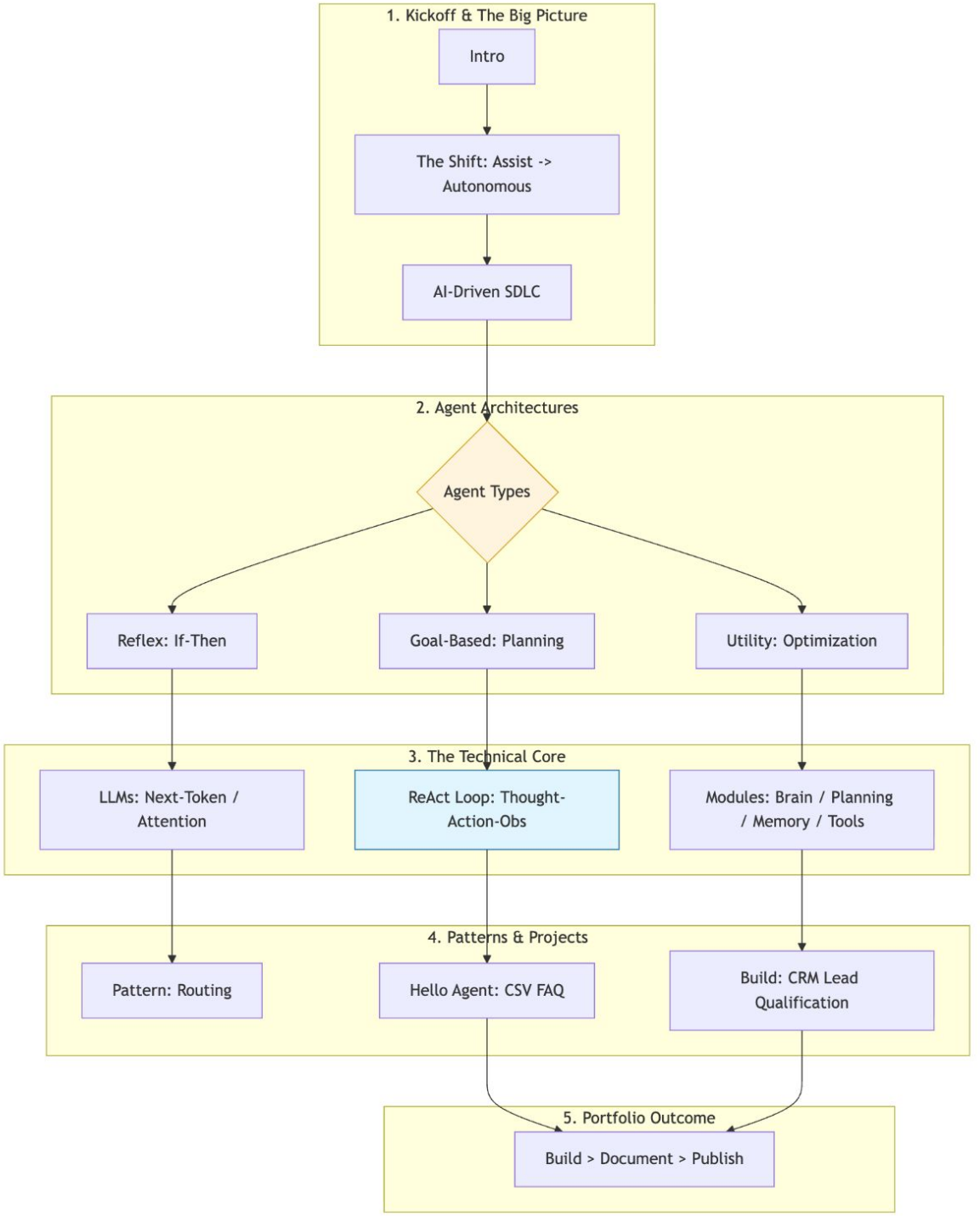
- This API key is for course-related usage only.
- Do not use it for personal projects or external experimentation.
- We have limited credits allocated per learner, and these are required for live sessions and assignment work.
- Keep your key secure and do not share it with anyone.

Looking forward to building powerful agentic systems with you.

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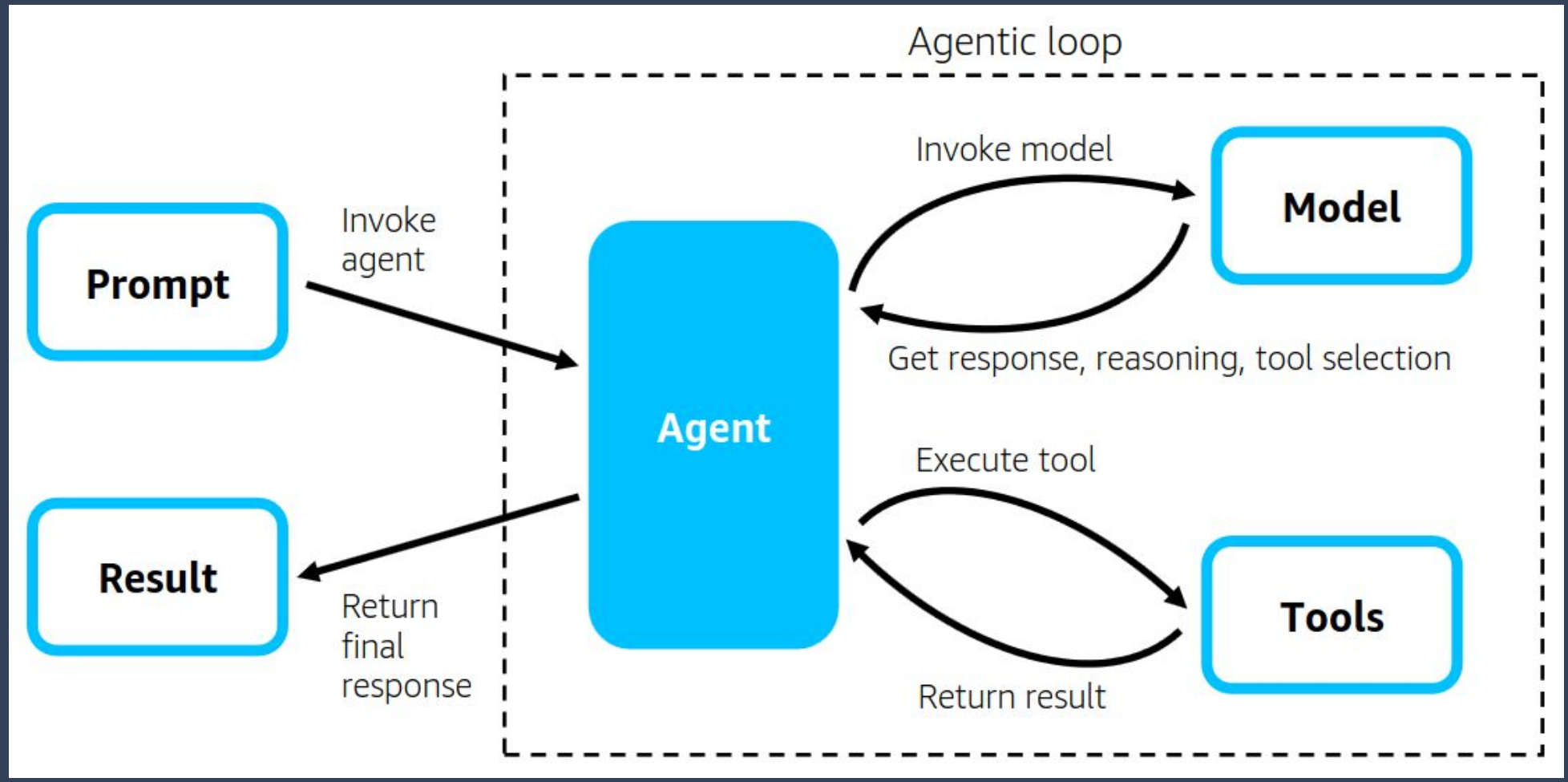
Week 1 Recap: Agentic AI Foundations

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Recap: The Anatomy of an Agentic loop



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Today's Agenda



1



Text Embeddings

2



Chunking Strategies

3



Indexing Strategies for RAG

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Let’s understand these concepts through a hands-on project

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Clone & Practice: Your Learning Repository



SupportDesk-RAG-Workshop

https://github.com/jeev1992/SupportDesk-RAG-Workshop

What's Included:

6 Complete Modules
Embeddings, Chunking, Indexing, RAG Pipeline, Evaluation, Agentic RAG

Working Code Examples
Ready-to-run demos, practice exercises, and solutions for each module

Getting Started

```
git clone https://github.com/jeev1992/
SupportDesk-RAG-Workshop.git
```

1. Install dependencies:
`pip install -r requirements.txt`
2. Configure OpenAI API key
3. Run demos from each module directory

Note: The following modules will be covered in **Week 3**: **RAG Pipeline, Evaluation, and Agentic RAG.**

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Problem statement: What exactly are we building?

In Week 2, we are building the retrieval layer for the SupportDesk RAG assistant, the part of the system that can reliably find the most relevant past support tickets for any new issue.

The problem we're solving (in the real world)

Engineers and Support teams and regularly face issues like:

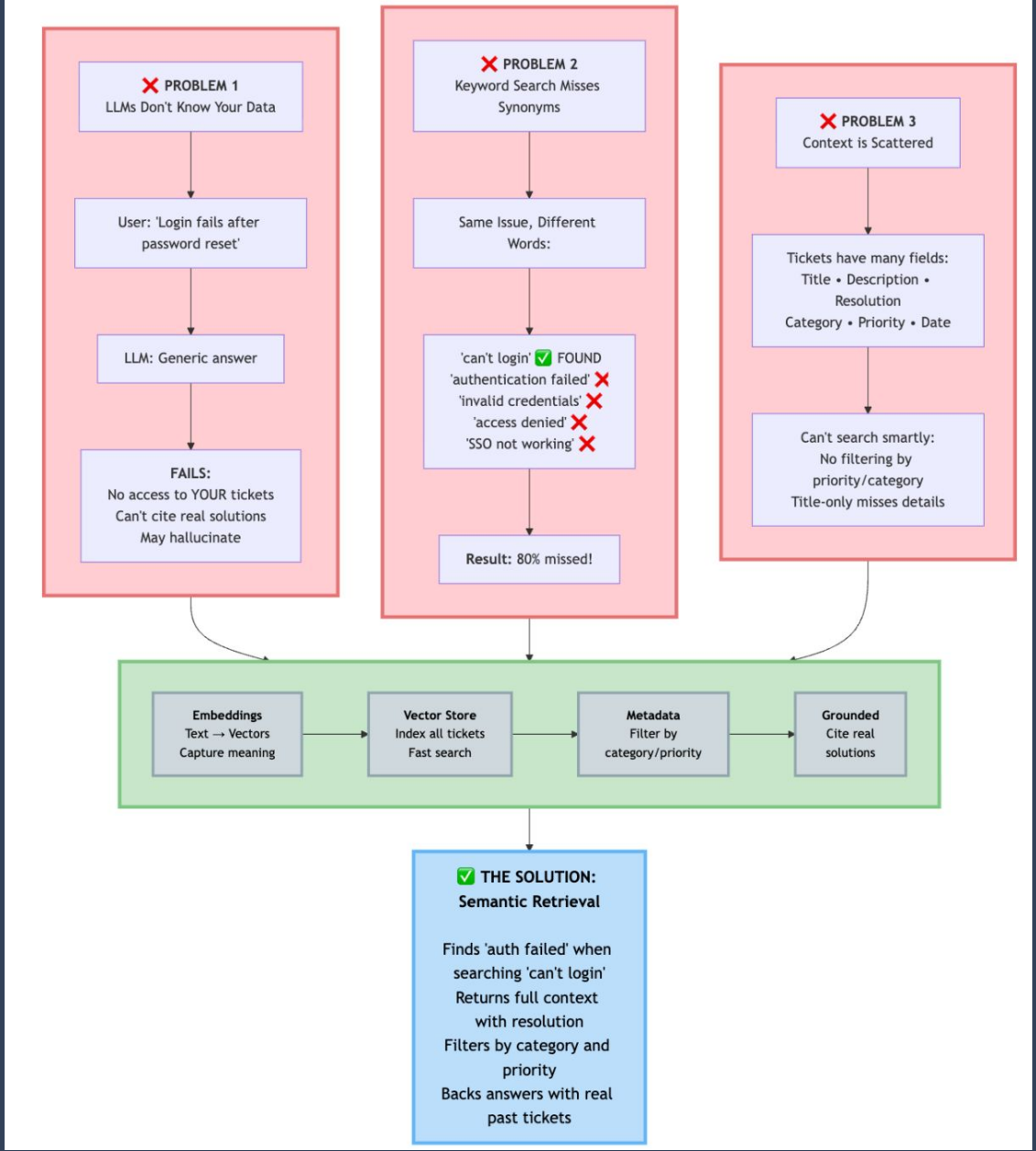
- Users can't log in after password reset
- Database connection timeouts in production
- Payments failing for international cards
- Emails not being delivered

The challenge is: **the solution often already exists in past tickets**, but finding it is slow because:

- Ticket data is **unstructured text** (not clean tables)
- People describe the same issue in **different words**
- Search is often **keyword-based**, so it misses relevant results
- There are too many tickets to read manually

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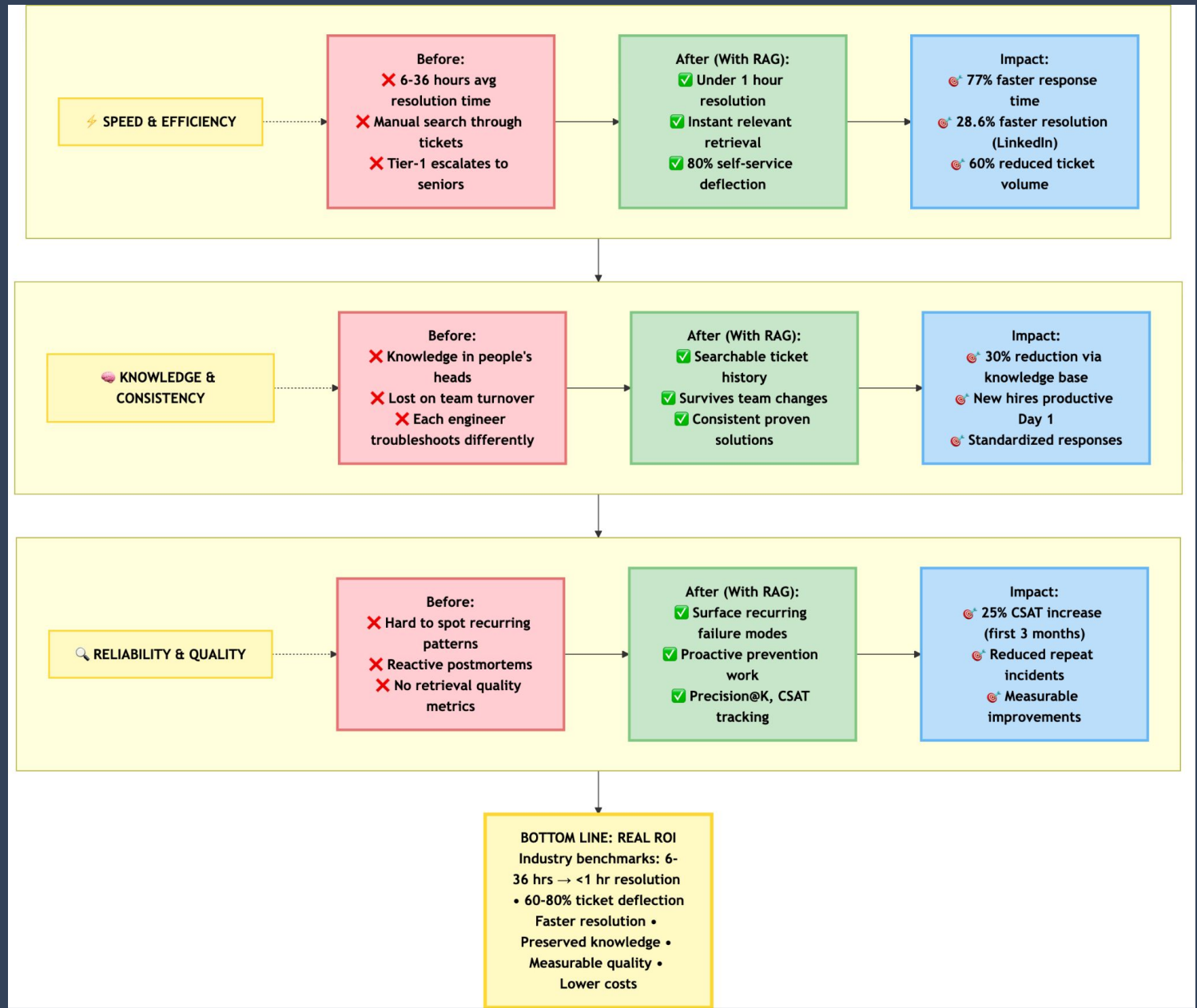
Why We Need Semantic Retrieval: The 3 Critical Gaps



- **LLM-only approach**: can sound correct but **may guess/hallucinate**, and it doesn't know your ticket history by default.
- **Context limits**: we can't paste all tickets into the prompt; we must retrieve only the best few.
- **Noise vs precision trade-off**: without chunking + indexing choices, retrieval becomes inconsistent:

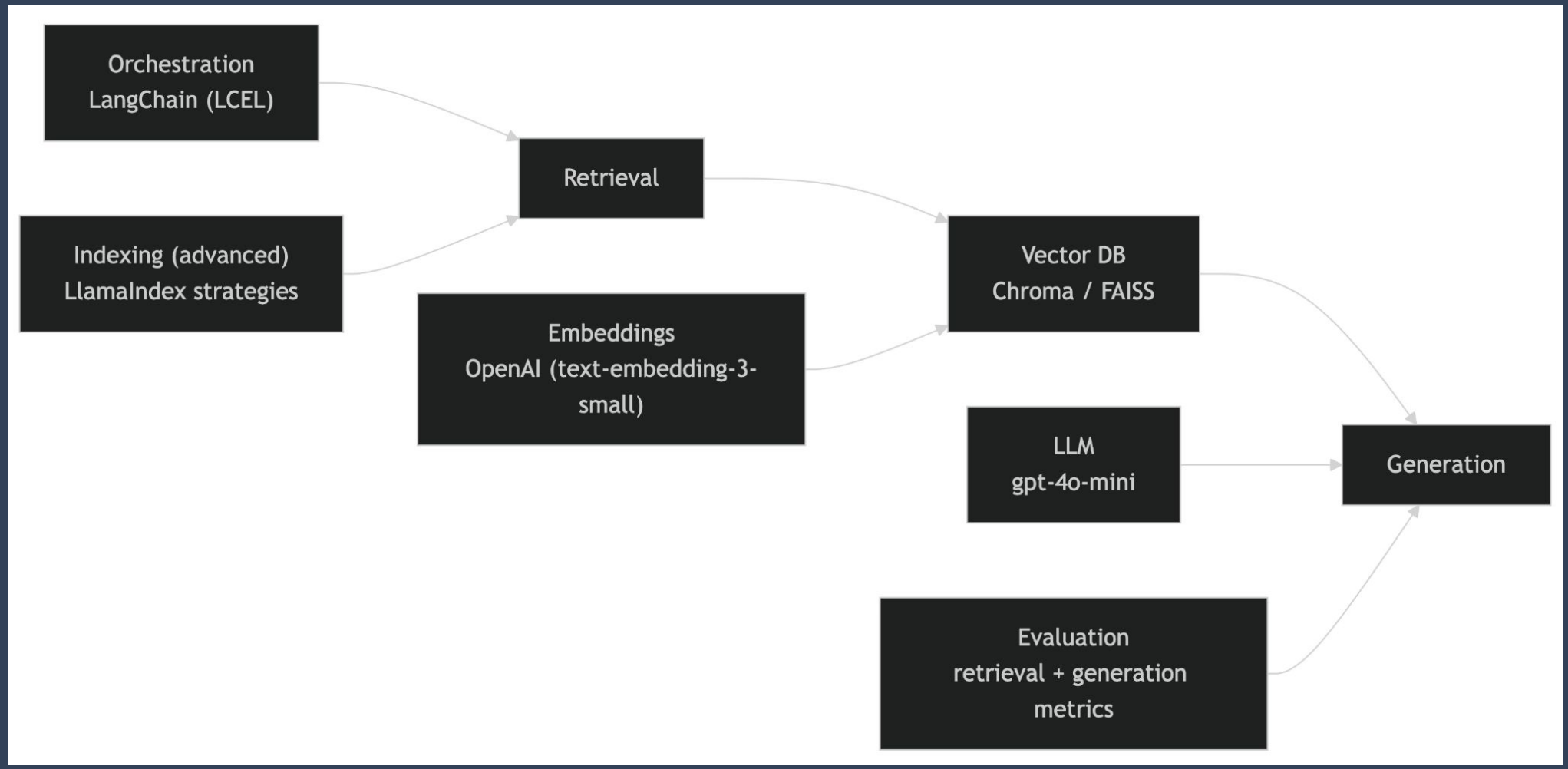
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Why this matters for Engineering Teams



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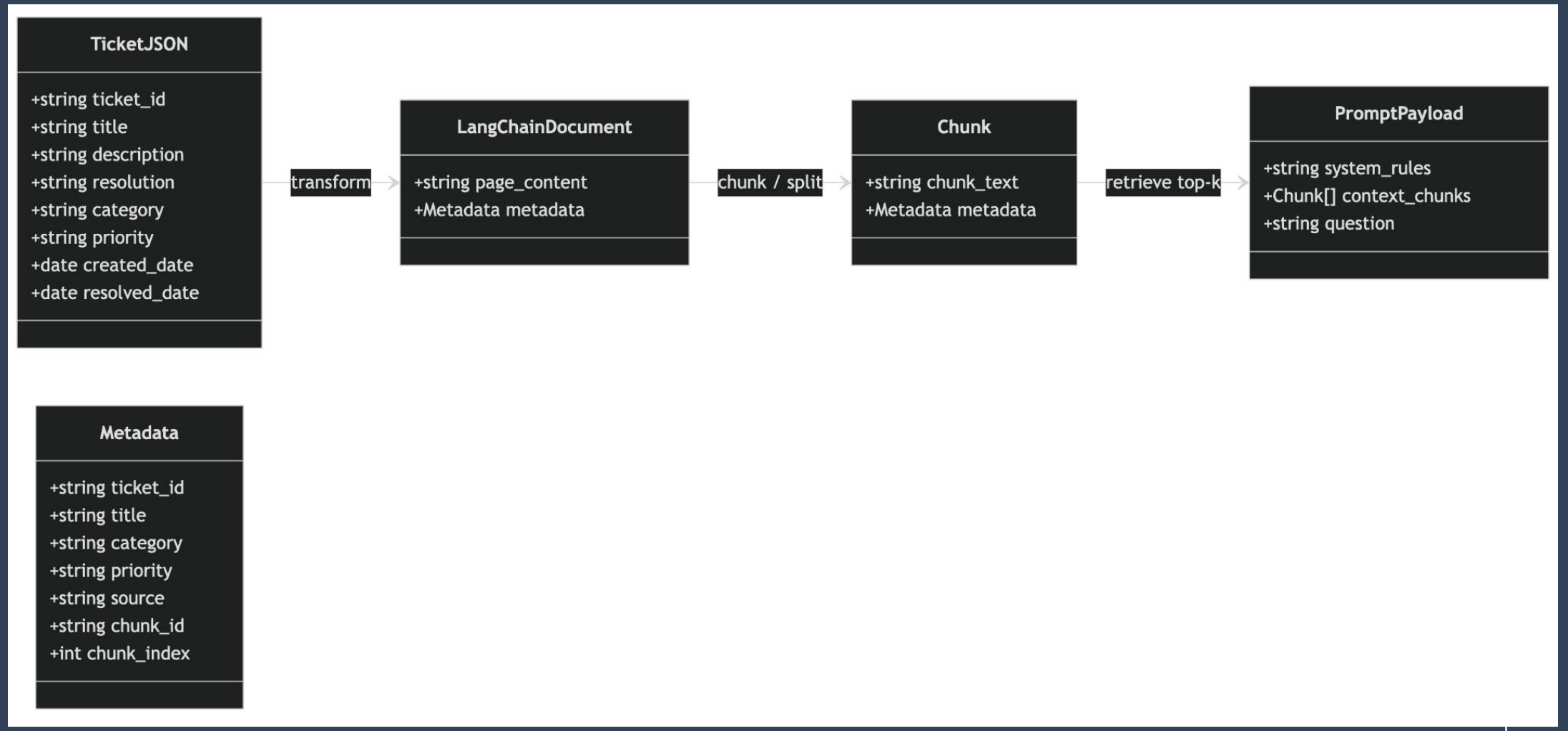
Key Tech Stack Across Weeks 2 and 3



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The Dataset

Data Schema (What the model sees):



Example Queries

What Users Will Ask	What the System Sees
Query 1 — Users can't log in after resetting their password.	**Target Match:** TICK-001 *(Authentication)* **Desired Output:** This is caused by stale session tokens. The fix is to clear active sessions and force re-authentication.
Query 2 — I'm seeing 402 errors on international payments.	**Target Match:** TICK-003 *(Payment Gateway)* **Desired Output:** Enable the international processing flag in the gateway configuration.
Query 3 — Why is the dashboard slow?	**Target Match:** TICK-010 *(Performance)* **Desired Output:** It is likely an N+1 query issue. Check if Redis caching is enabled for aggregate stats.

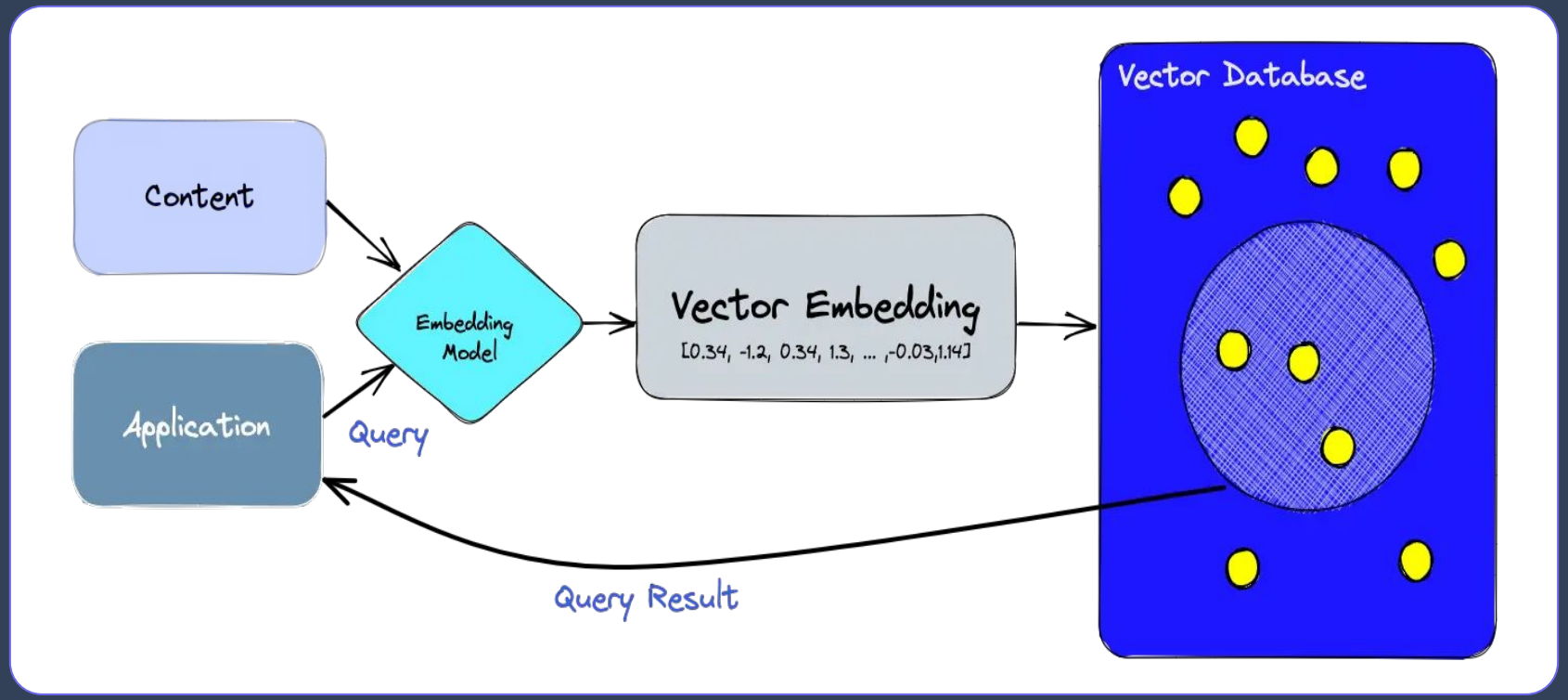
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Let's Get It Started

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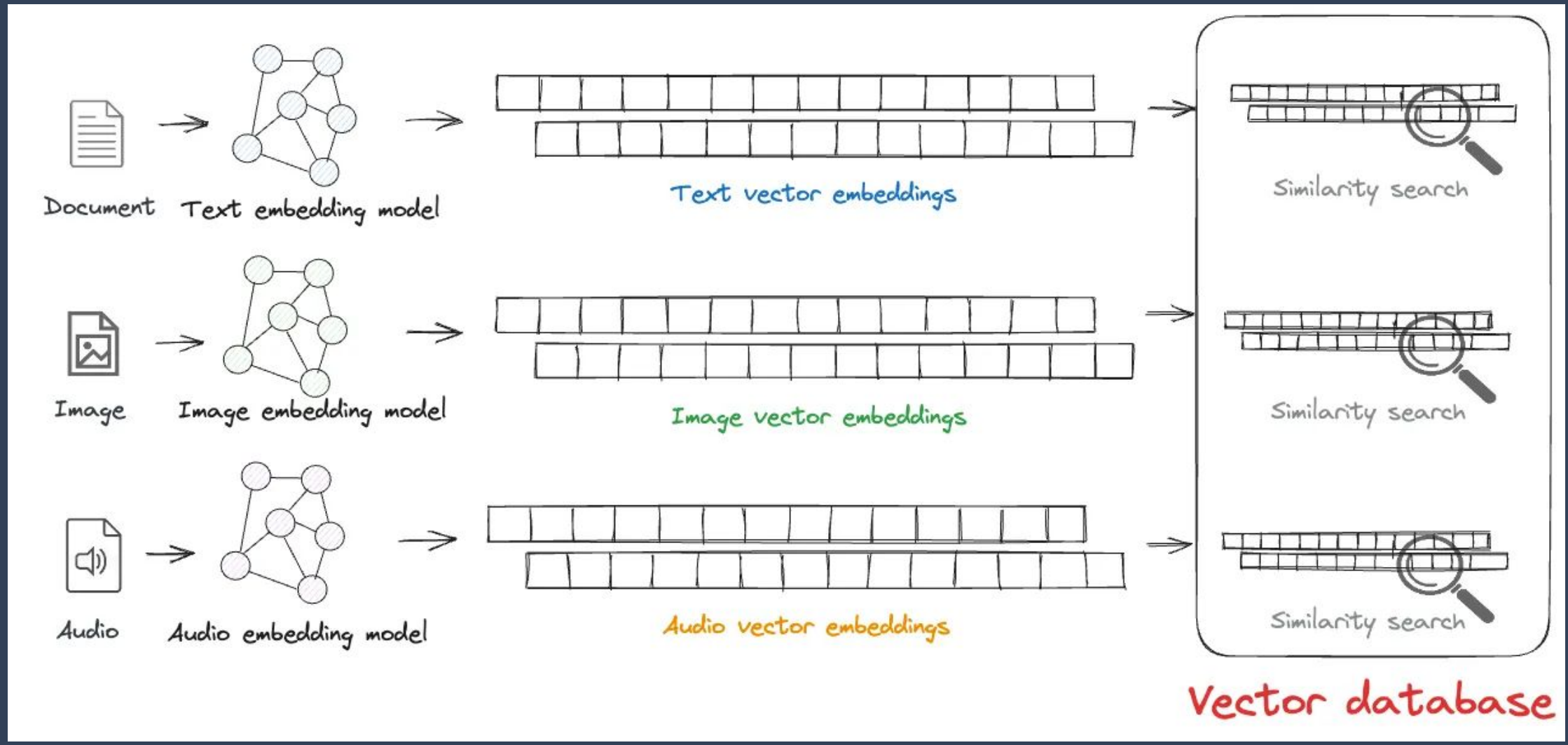
Text Embeddings: Understanding Text Embeddings

Vector Databases - Why you should know it?



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Storing the embeddings in vector databases



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Vector Store

- We know how to search for data in DB (SQL or NoSQL)
- But how to search for text data?





- **Can we do string match?**

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Vector Store

- We know how to search for data in DB (SQL or NoSQL)
- But how to search for text data?





- **Can we do string match? - Why not**

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Vector Store

Can we do string match?

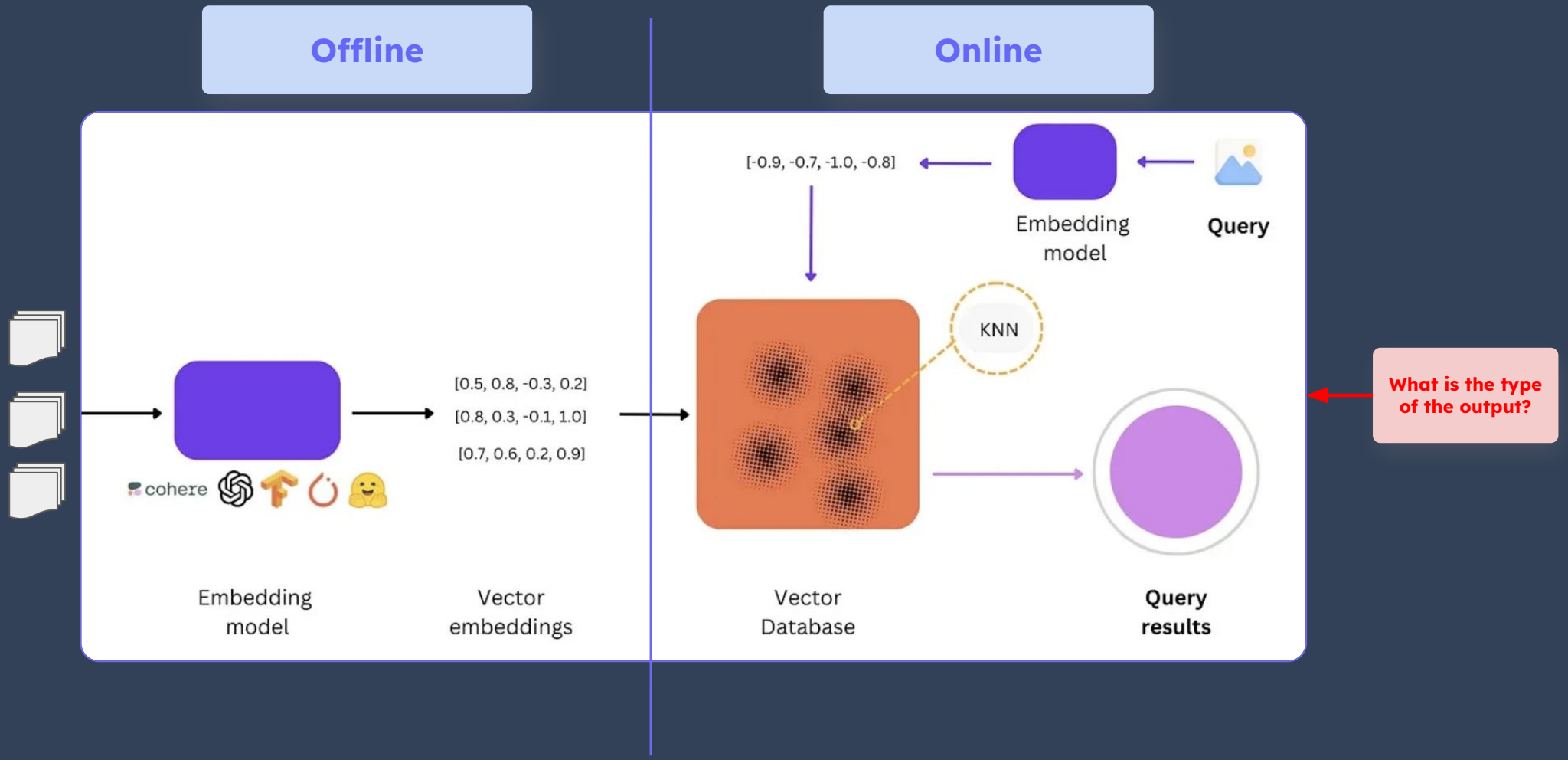
- Does not handle synonyms or rephrasing
  - **“Car”** vs **“Vehicle”**

- No semantic understanding
  - **“I am travelling to Mexico, what phone plan should I use”** vs **“International plans”**

What can we use instead?

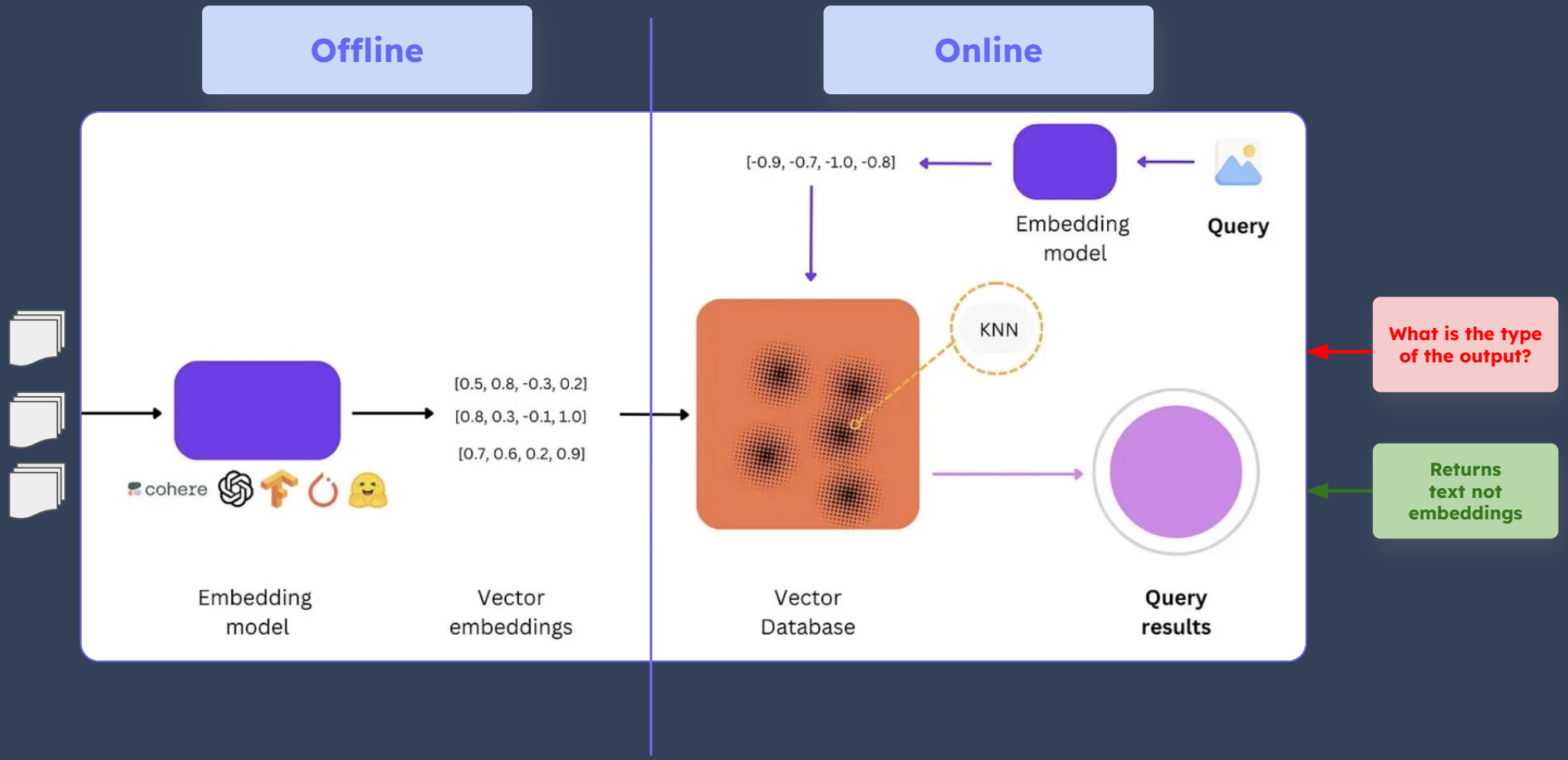
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Vector Store



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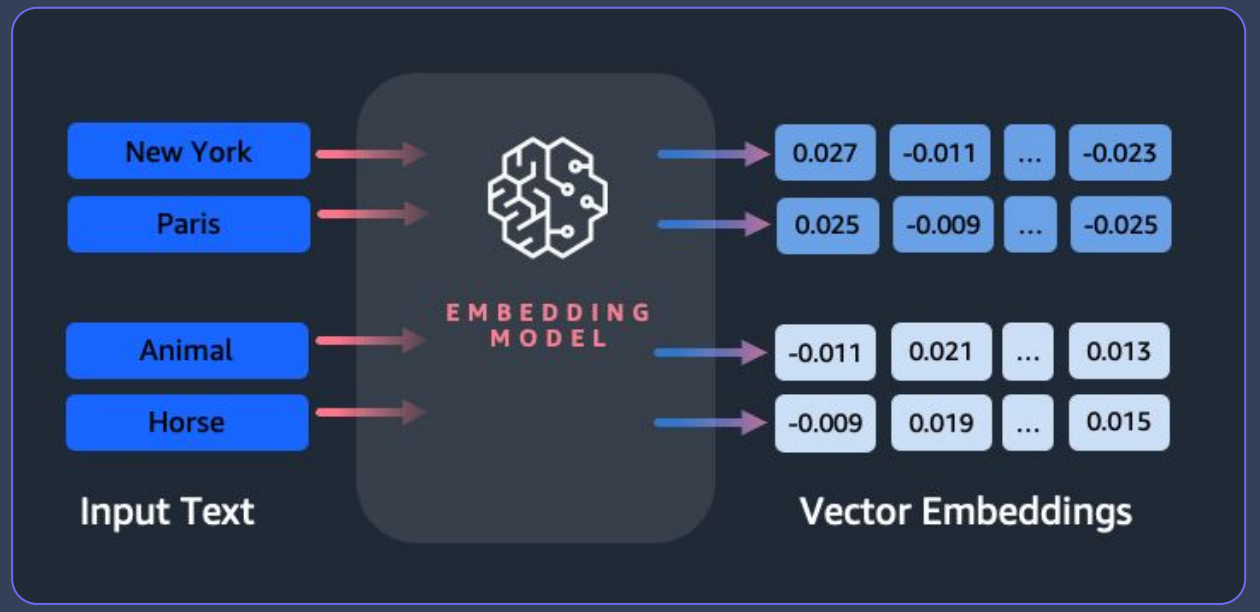
Vector Store



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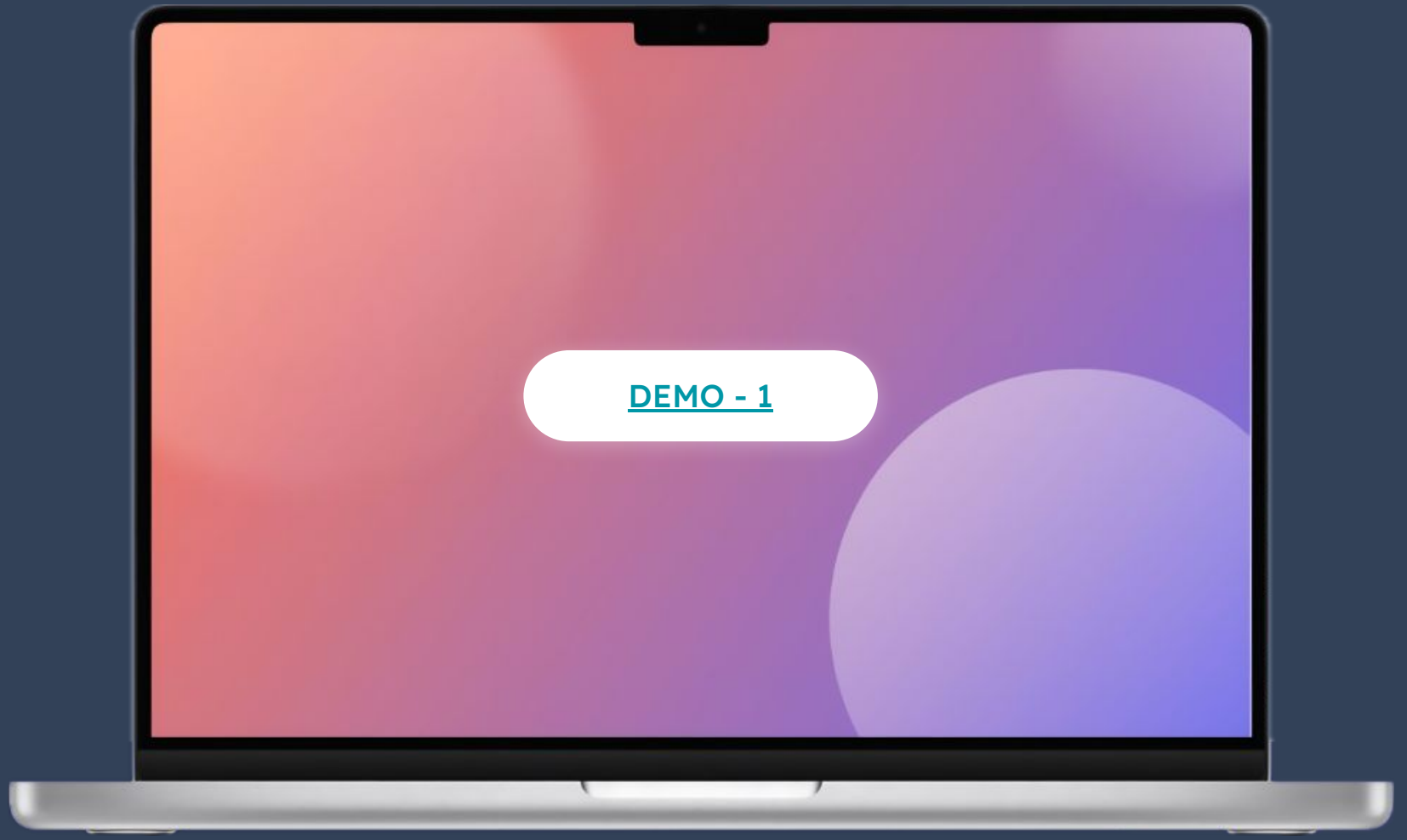
What is Embedding?

- Embeddings create a **vector representation** of a piece of text.
- We can think about text in the vector space
- Do things like semantic search
  - Where we look for pieces of text that are most similar in the vector space.



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Module 1 - Implementation Notes: Text Embeddings



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No Question Is Too Small Every Question Deserves an Answer!!

Ask about anything:

- Concepts from today's modules
- How this applies to your work
- Implementation details
- Clarifications on what we covered

Quick Request

- Please do not mind if a question gets skipped accidentally
- Repost politely in chat if needed
- We will try our best to cover as many questions as possible

Remember: If you're wondering about it, others probably are too.



Time for a Break



A 10-minute break is starting now. Please return when the timer ends to continue the presentation

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Today's Agenda



1

Text Embeddings



2

Chunking Strategies



3

Indexing Strategies for RAG

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Chunking: Breaking Down Long Documents



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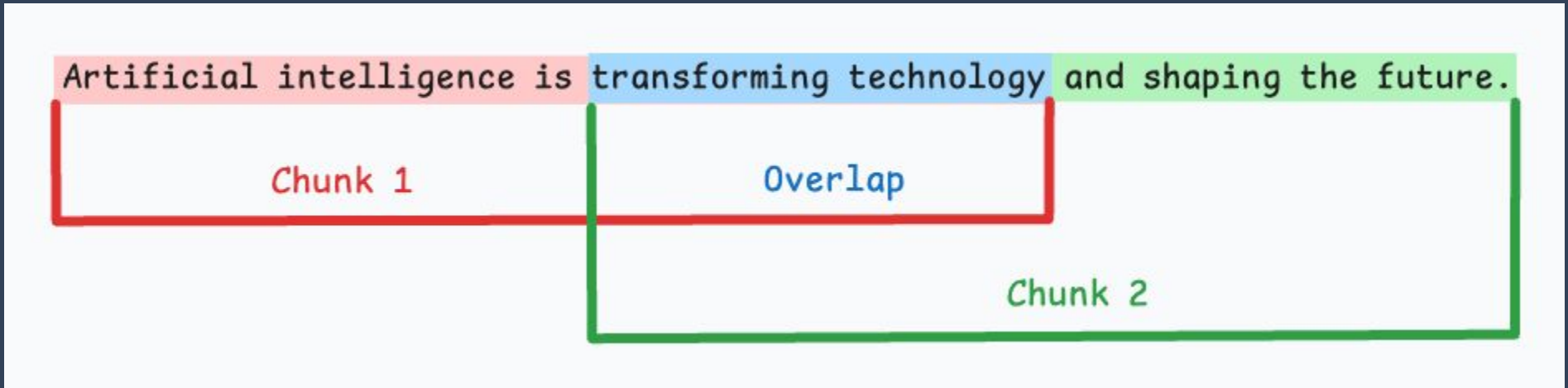
Chunking: Breaking Down Long Documents



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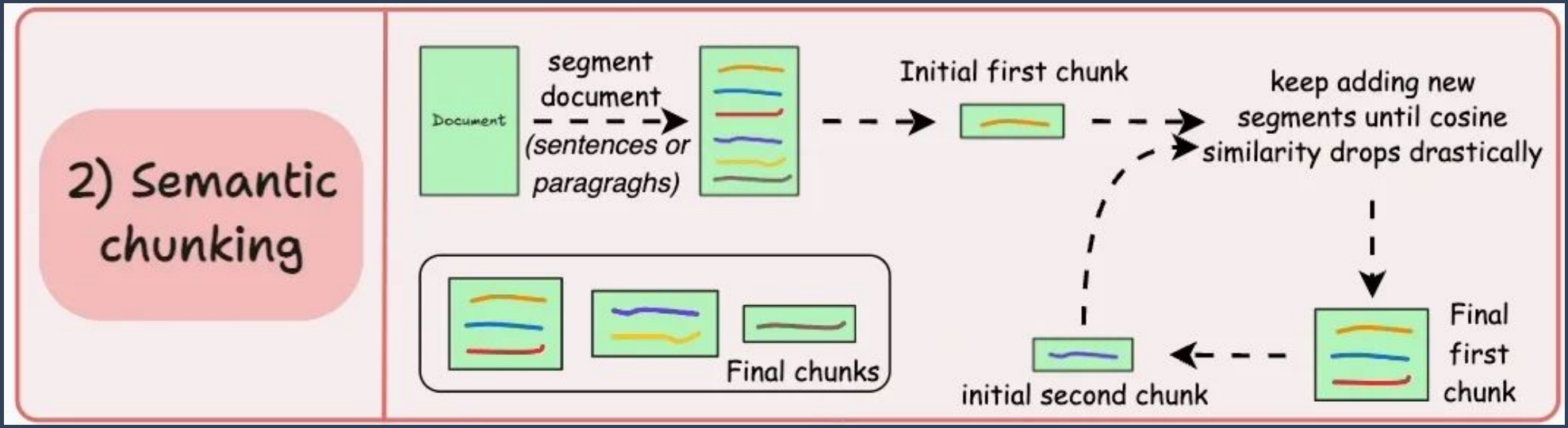
1) Fixed-size chunking

The most intuitive and straightforward way to generate chunks is by splitting the text into uniform segments based on a pre-defined number of characters, words, or tokens.



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2) Semantic chunking



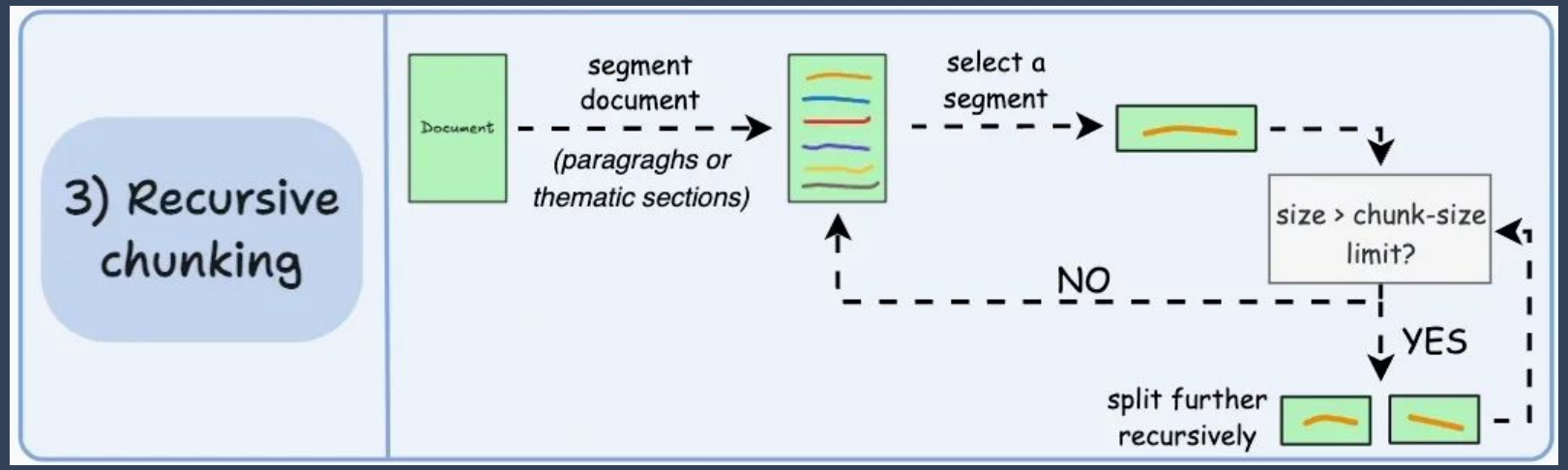
- Segment the document based on meaningful units like sentences, paragraphs, or thematic sections.
- Next, create embeddings for each segment.

Here's what the output could look like:

```
Artificial intelligence is transforming industries by automating
processes, enhancing decision-making, and providing insights through
data analysis. Machine learning, a subset of AI, enables systems to learn
and improve from experience without explicit programming. Deep learning,
a branch of machine learning, uses neural networks with multiple layers
to model complex patterns in data.
```

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3) Recursive chunking



First, chunk based on inherent separators like paragraphs, or sections. Next, split each chunk into smaller chunks if the size exceeds a pre-defined chunk size limit. If, however, the chunk fits the chunk-size limit, no further splitting is done.

Here's what the output could look like:

Paragraph 1

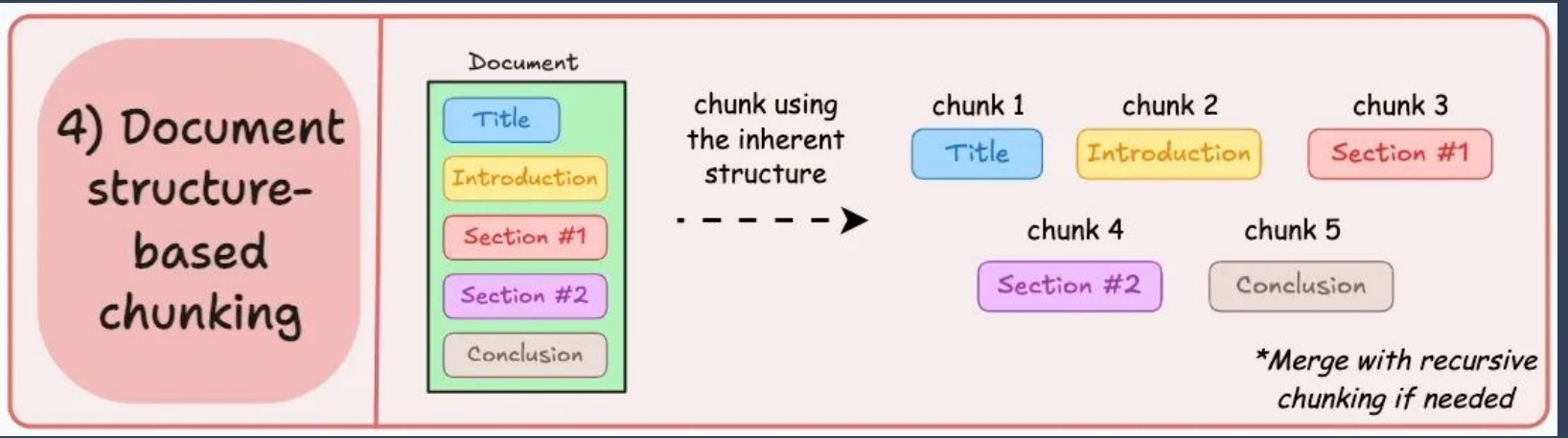
Artificial intelligence is transforming industries by automating processes, enhancing decision-making, and providing insights through data analysis. Machine learning, a subset of AI, enables systems to learn and improve from experience without explicit programming. Deep learning, a branch of machine learning, uses neural networks with multiple layers to model complex patterns in data.

Paragraph 2

AI is also improving natural language processing, enabling applications like chatbots and virtual assistants.

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4) Document structure-based chunking



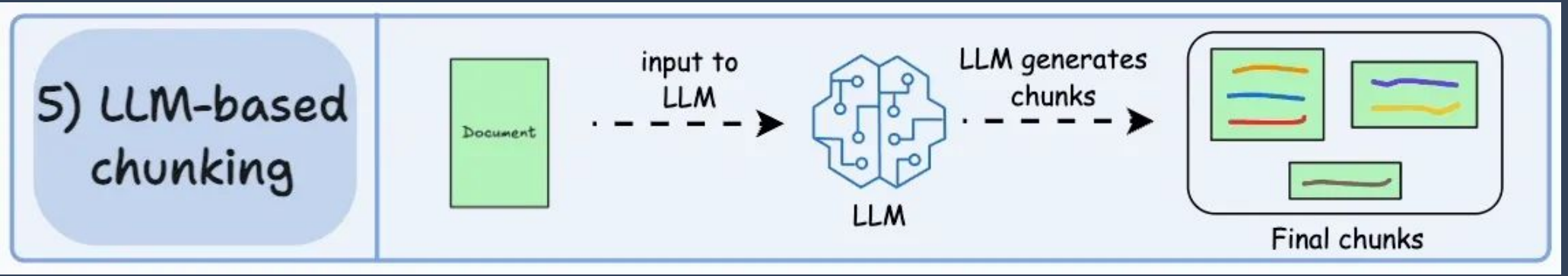
It utilizes the inherent structure of documents, like headings, sections, or paragraphs, to define chunk boundaries. This way, it maintains structural integrity by aligning with the document's logical sections.

Here's what the output could look like:

Title: The Role of Artificial Intelligence in Modern Education	Chunk 1
Introduction Artificial intelligence (AI) is reshaping education by providing personalized learning experiences and automating administrative tasks.	Chunk 2
Section 1: Personalized Learning AI enables the customization of educational content to meet individual student needs, enhancing engagement and comprehension.	Chunk 3
Section 2: Administrative Automation From grading to scheduling, AI tools are streamlining administrative processes, allowing educators to focus more on teaching.	Chunk 4
Conclusion The integration of AI in education holds the promise of more efficient learning environments and improved student outcomes.	Chunk 5

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5) LLM-based chunking

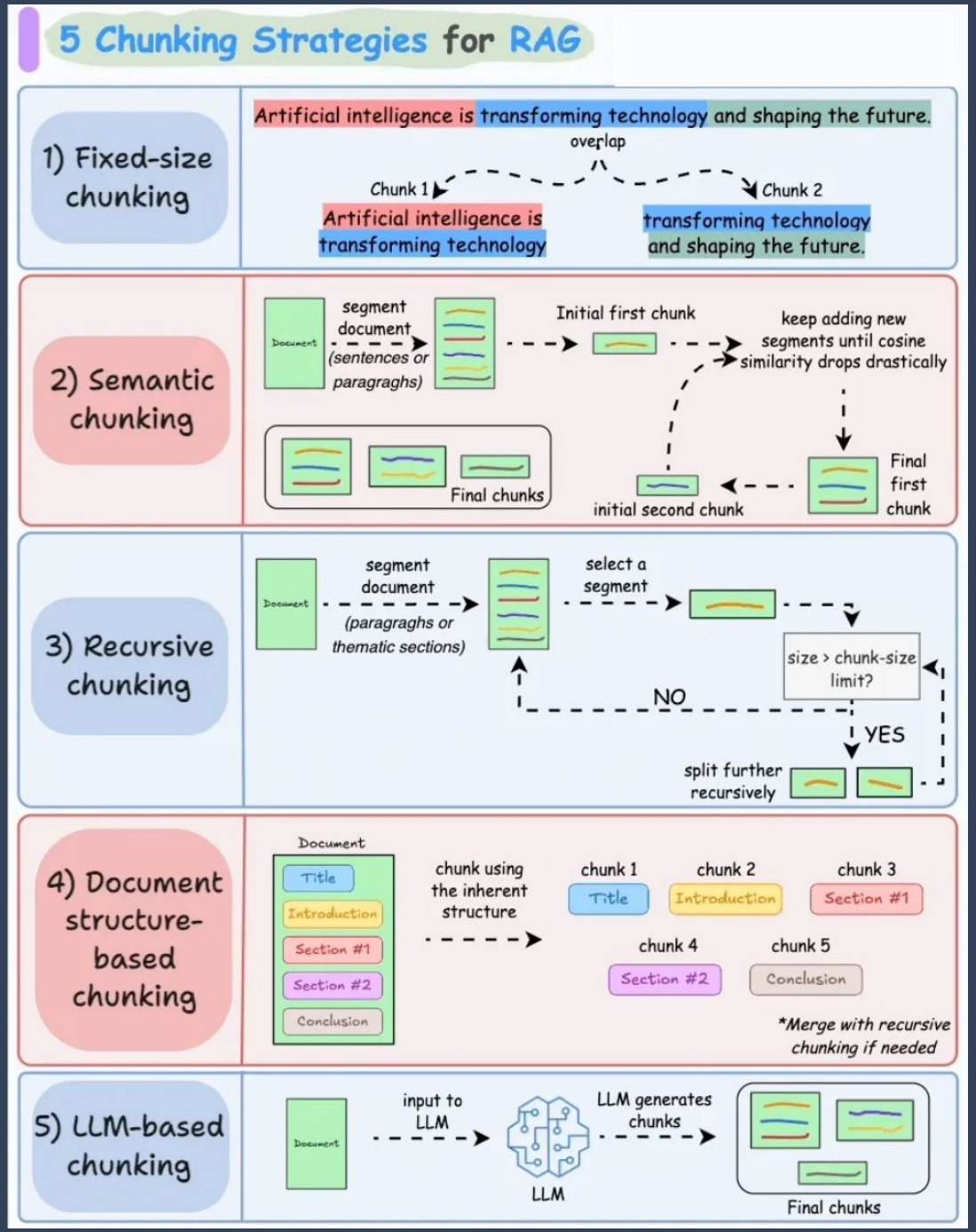


- Since every approach has upsides and downsides, why not use the LLM to create chunks?
- The LLM can be prompted to generate semantically isolated and meaningful chunks.
- Quite evidently, this method will ensure high semantic accuracy since the LLM can understand context and meaning beyond simple heuristics (used in the above four approaches).

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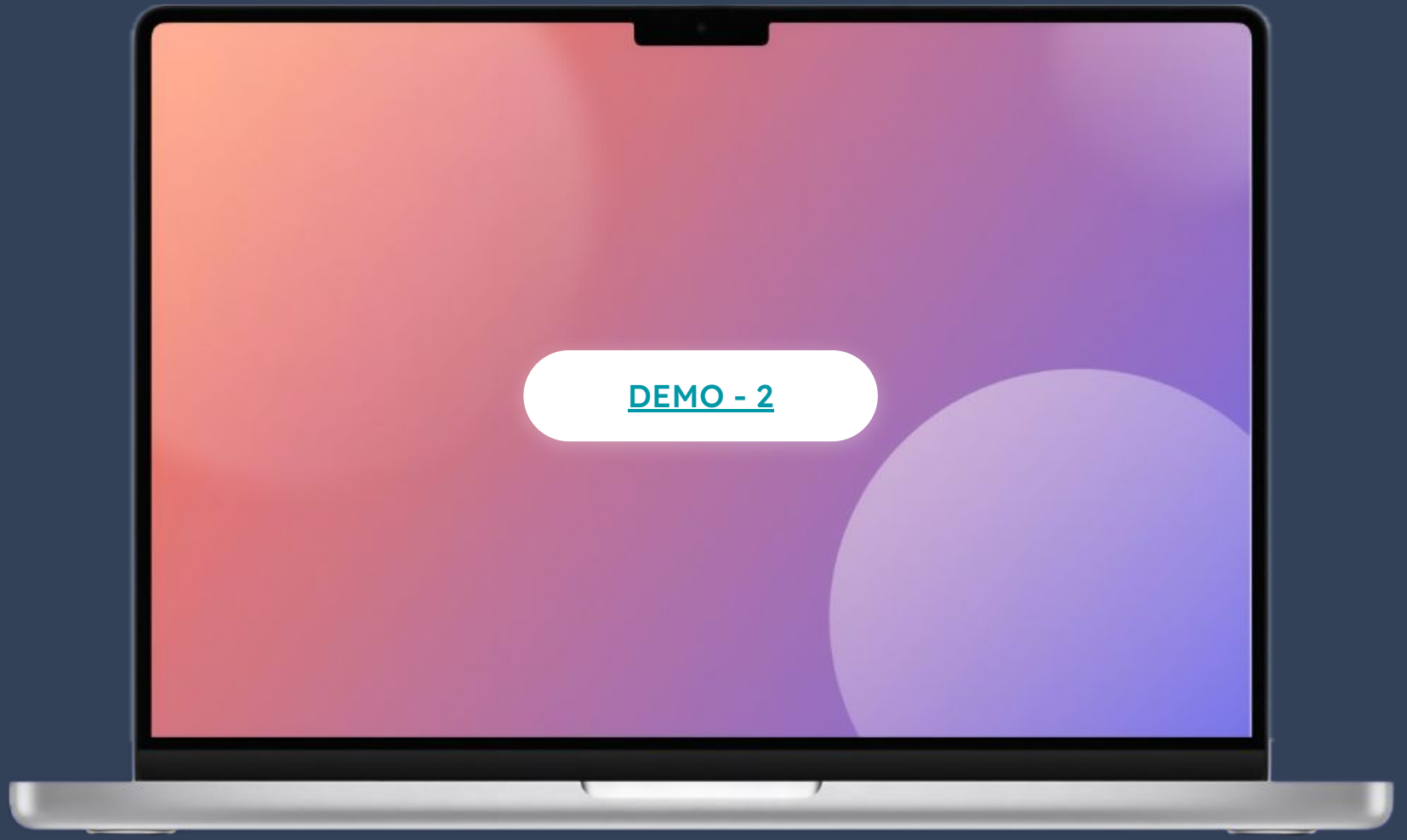
Cheat Sheet : 5 chunking strategies for RAG

👉 Refer back to this slide during revision to choose the right chunking strategy



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Module 2 - Implementation Notes: Chunking Strategies



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No Question Is Too Small Every Question Deserves an Answer!!

Ask about anything:

- Concepts from today's modules
- How this applies to your work
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- Clarifications on what we covered

Quick Request

- Please do not mind if a question gets skipped accidentally
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Remember: If you're wondering about it, others probably are too.



Live Zoom Poll



We are launching a poll on Zoom. Please take 2 minutes to fill out the form.

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Time for a Break



A 5-minute break is starting now. Please return when the timer ends to continue the presentation

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Today's Agenda

1



Text Embeddings

2



Chunking Strategies

3

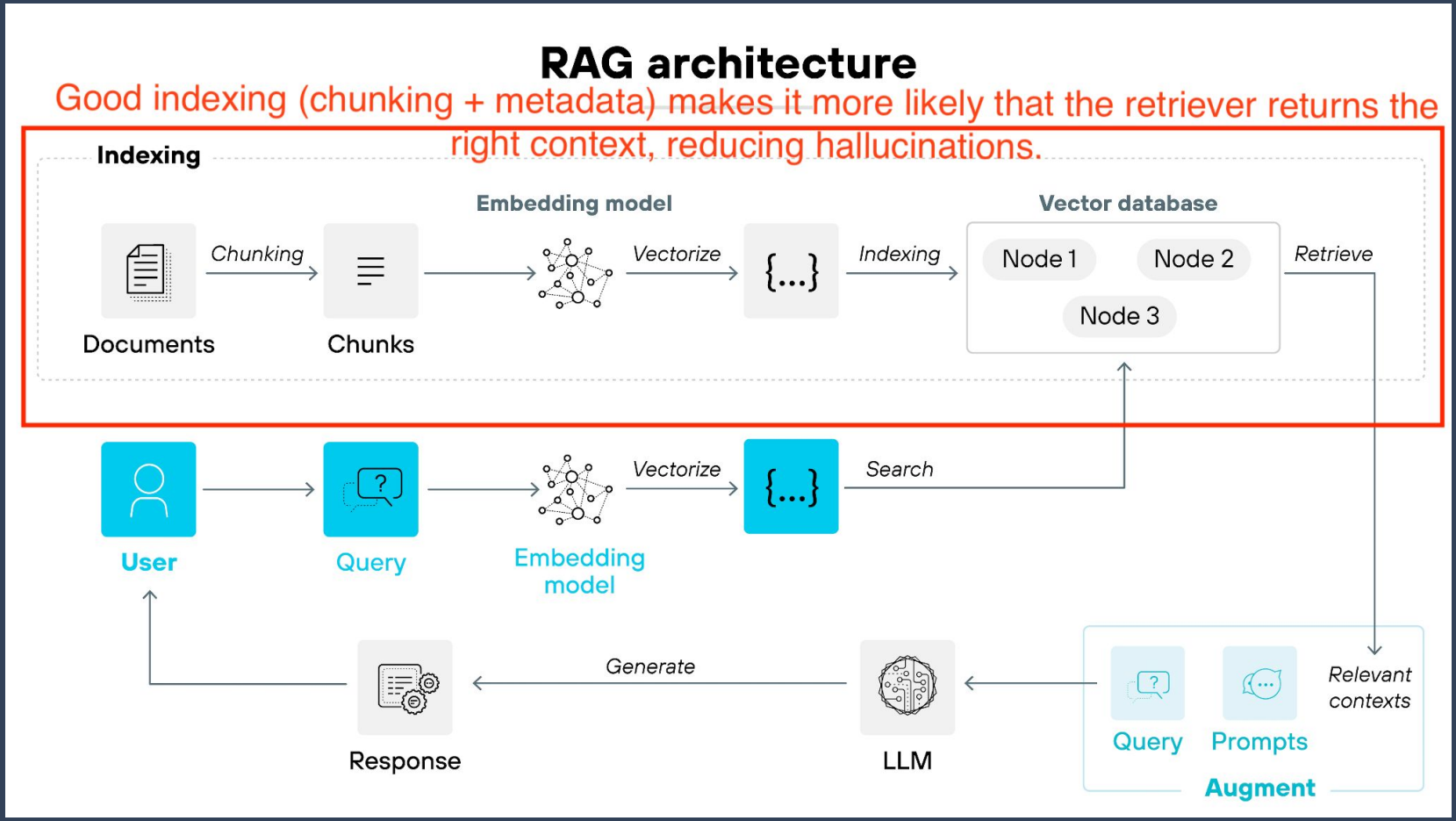


Indexing Strategies for RAG

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Why Indexing is Required in RAG

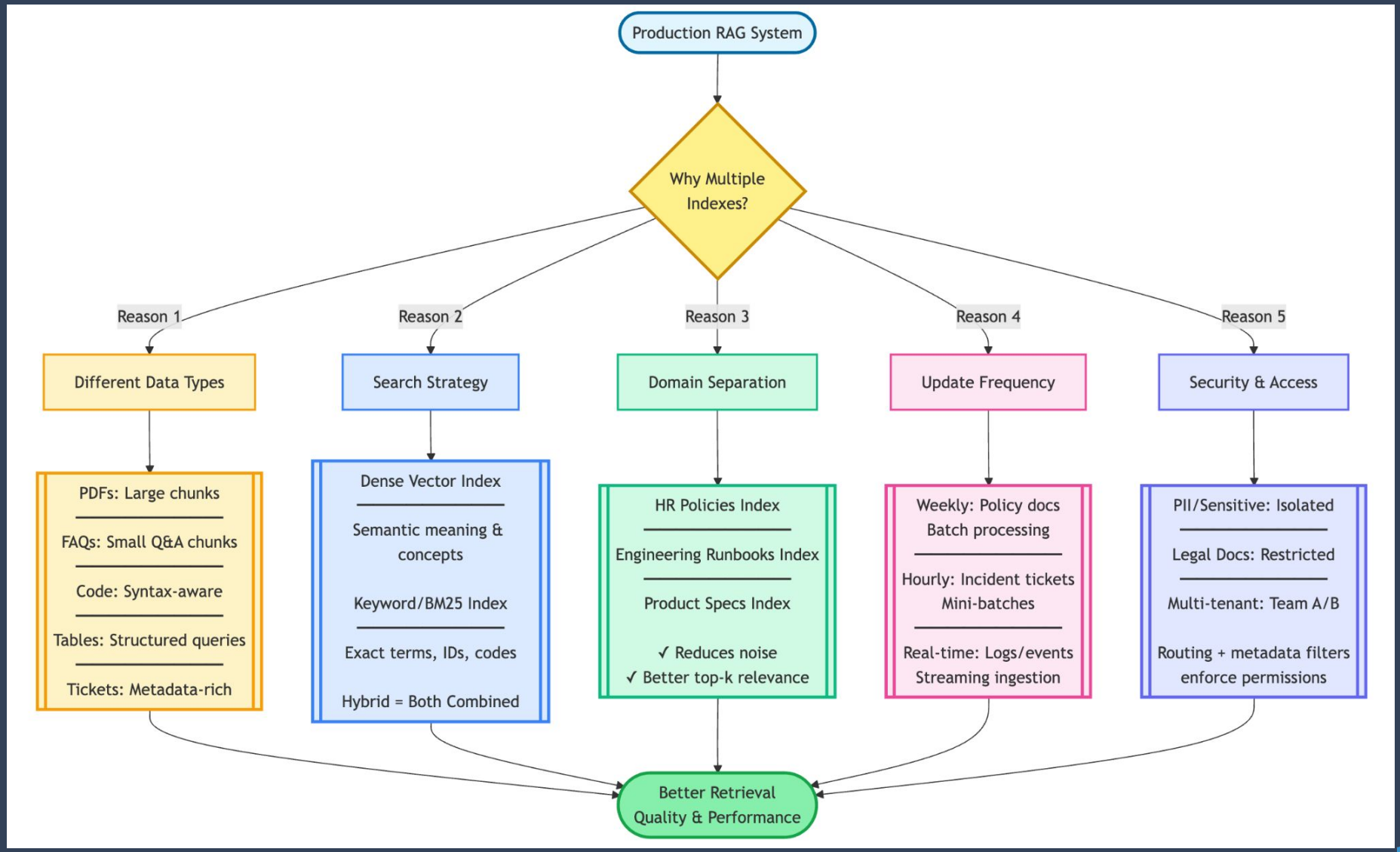
- **RAG needs a fast way to find relevant evidence.** Indexing converts documents into **retrievable units (chunks/nodes)** with embeddings + metadata.
- **Without indexing, retrieval is too slow at scale.** You'd be doing a near **full scan** over all chunks every query (high latency, high cost).



Indexing happens **offline**; retrieval happens **online at query time**

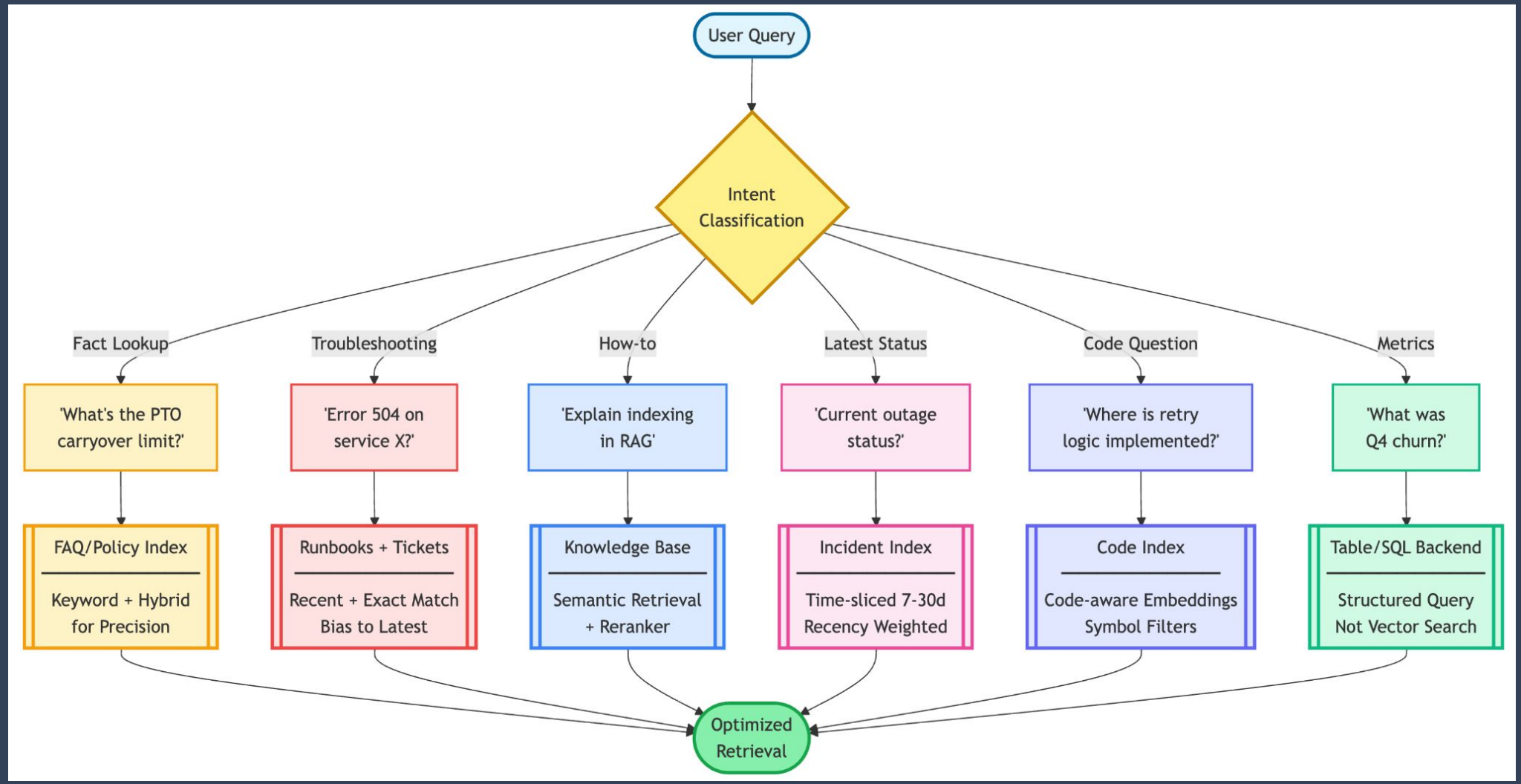
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Why multiple indexes exist in production RAG



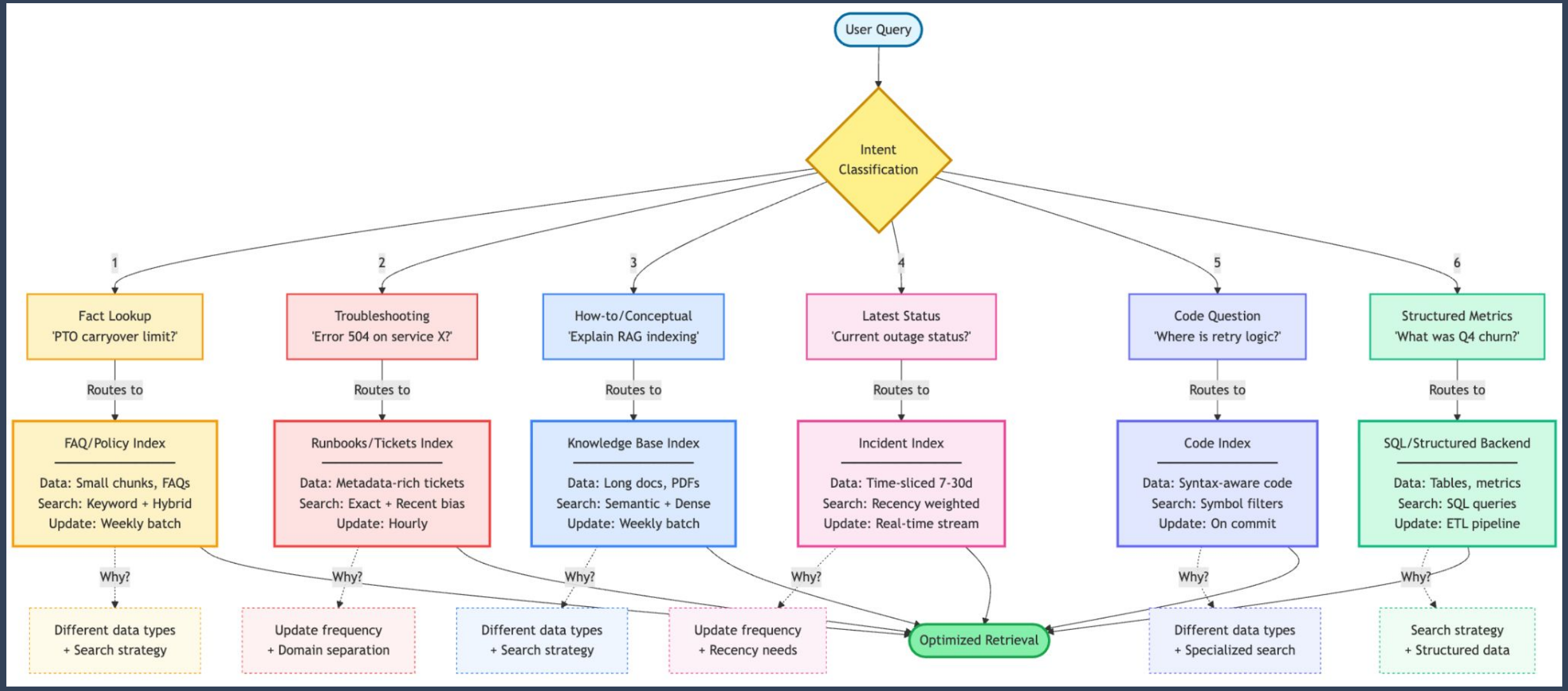
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Why route queries differently based on intent



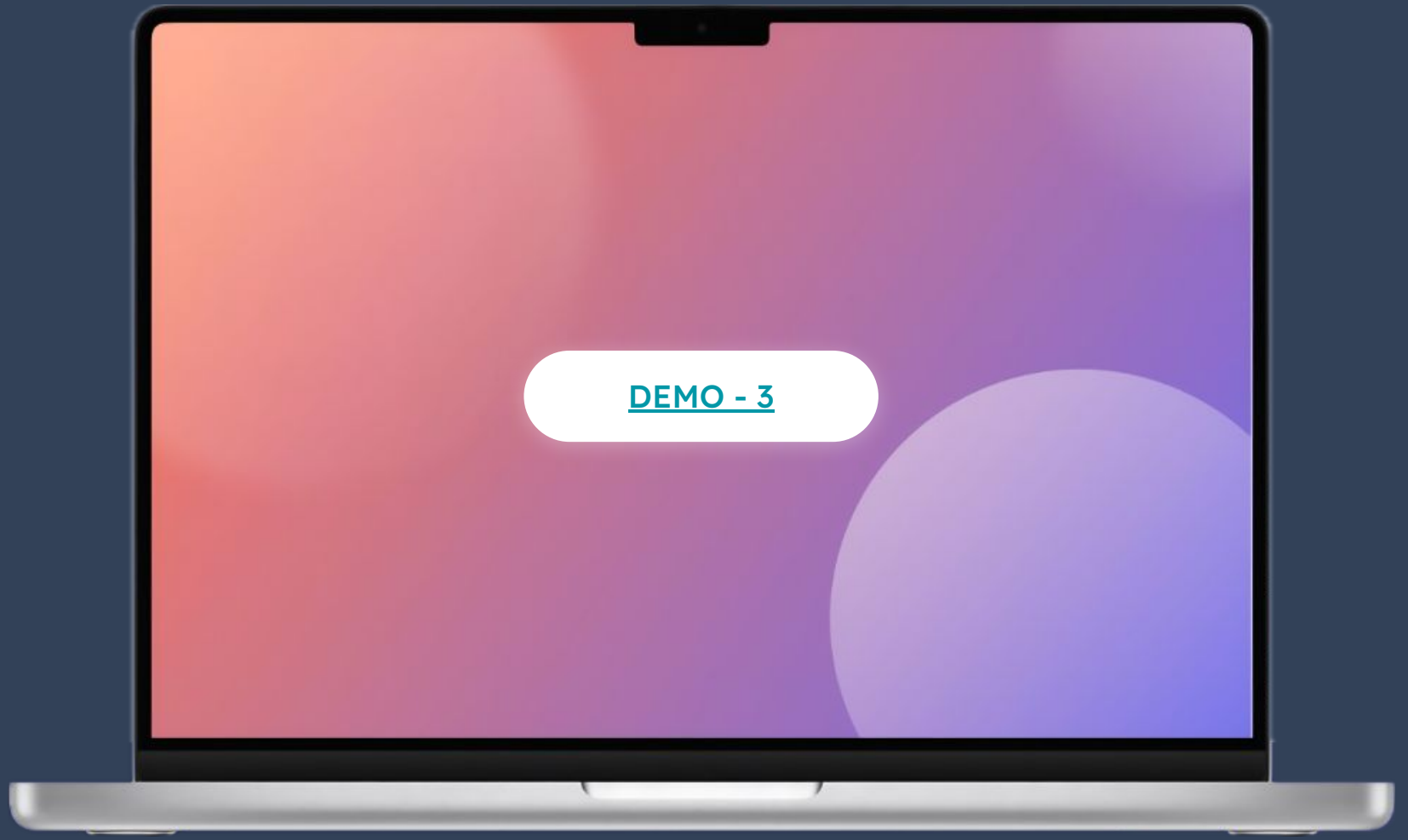
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Intent-Based Query Routing in Production RAG Systems



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Module 3 - Implementation Notes: Indexing Strategies for RAG



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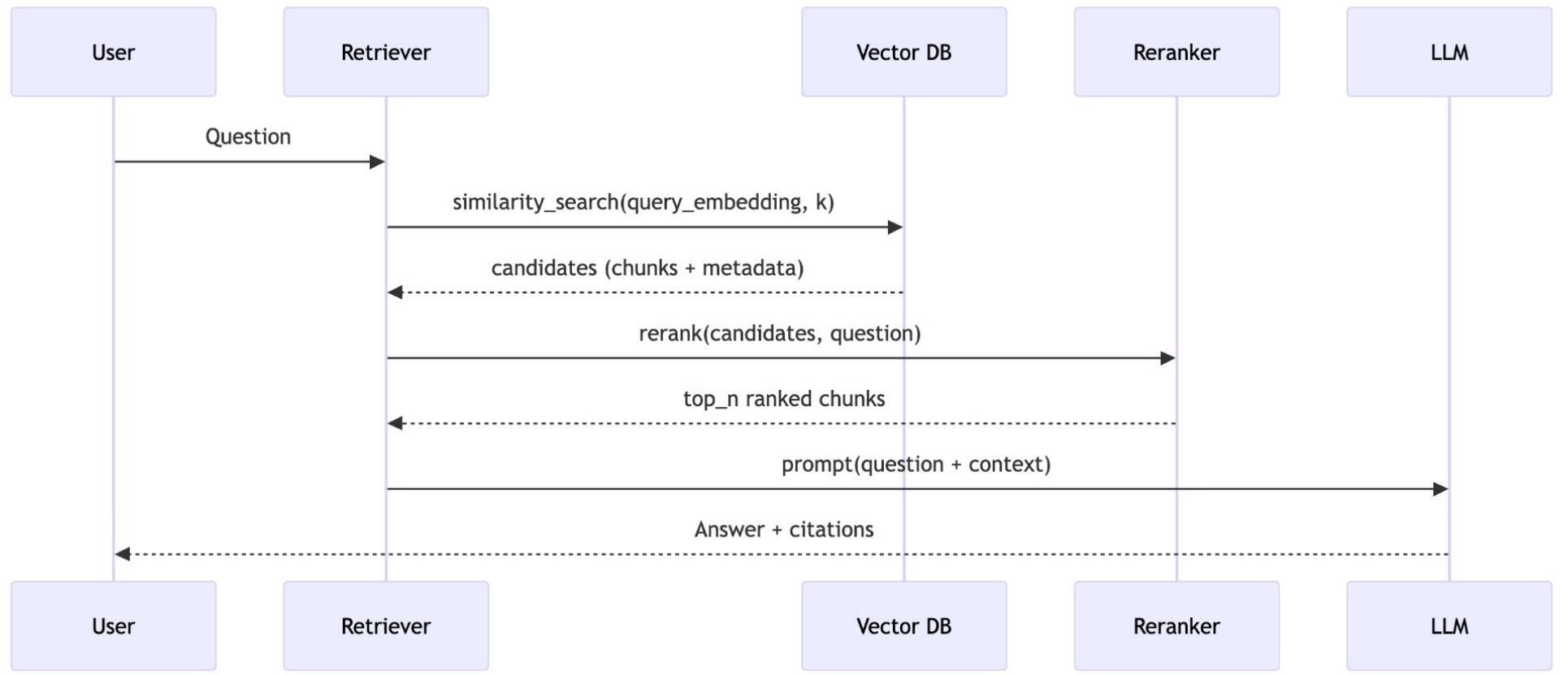
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Retrieval in One Picture (The Core Loop)

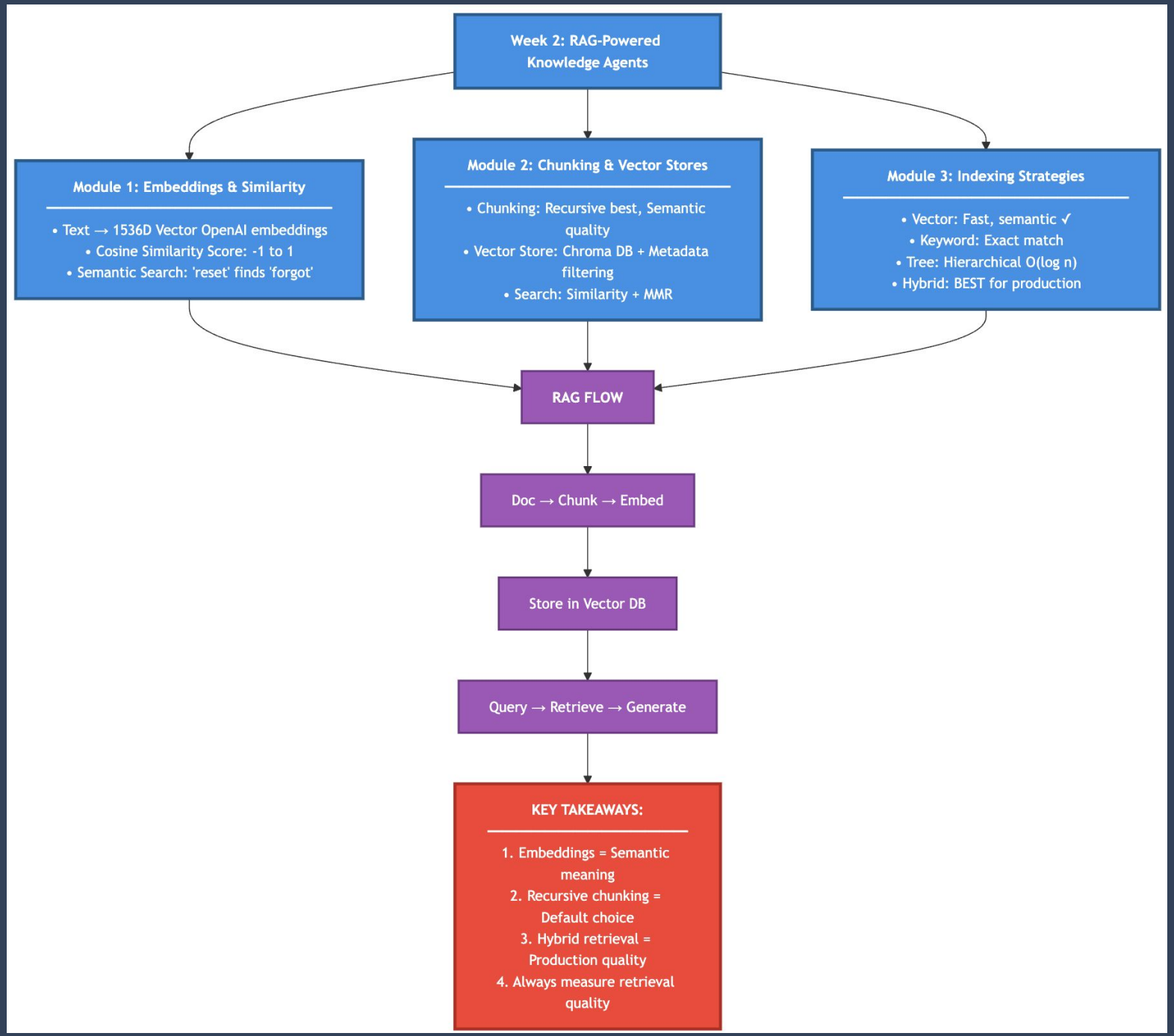
- Convert question → searchable representation
- Fetch candidates → Top-K chunks
- Improve relevance → rerank
- Send best context → grounded answer



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Week 2 Summary: From Embeddings → Indexing

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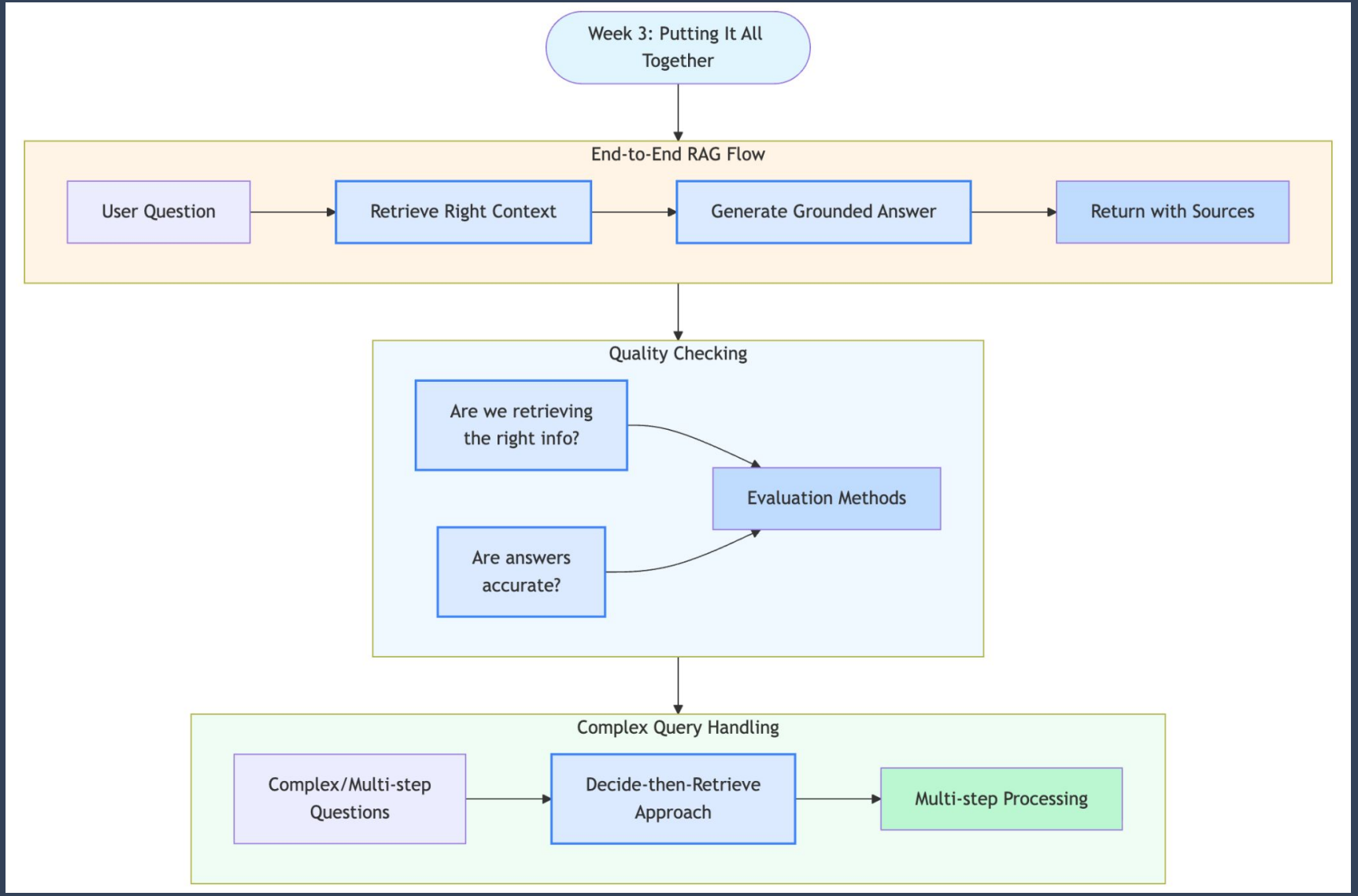


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Glimpse of Week 3



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